I2C bus A (IMU + actuator ADS1115 0x48/0x49): SDA=41 SCL=42
I2C bus B (power ADS1115 0x48/0x49/0x4A, TCA9535 0x20, ext): SDA=19 SCL=20 (USB-OTG port unusable)
ACT_EN=39 (through E-stop loop) ESTOP_SENSE=8 PV_EN=40 IMU_INT=3 LED_ERR=46
TCA9535: P00..P03=BAT1..4_EN P04=IMU_RST P05=PWR LED P06..P17=EXP_IO0..9

Power: up to 4 packs (populated: BAT1, BAT2) 10S..20S + PV, all channels rated 100 V
(FETs −100 V: for 20S use PV Voc <= 90 V; TVS SMCJ58A is for 10S – use SMCJ85A for 20S).
Pack fuses are external (fit one per pack, close to the pack).

power_path

Power path: battery & PV channels, bus

File: power_path.kicad_sch

supply_sense

Logic supply, power ADCs, IO expander

File: supply_sense.kicad_sch

mcu

ESP32-S3, IMU, E-stop

File: mcu.kicad_sch

actuators

Linear actuators: 6x IBT-2, position & current sense

File: actuators.kicad_sch

Root sheet
Main controller – ESP32-S3
solbot

Sheet: /
File: solbot_main.kicad_sch

Title: solbot_main v4 – ESP32-S3 main controller

Size: A3	Date: 2026-10-15	**Rev: v4.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

BAT1

BAT1 channel: 1 mOhm Kelvin shunt, INA241A2 (20 V/V, REF=Vs/2 -> bidirectional, +-82 A FS), back-to-back P-FET switch. Gate driver sinks ~2 mA via MMBTA42. 12 V zener sets Vgs independent of bus voltage (105..205).

J1 BAT1+ (M4 ring) WSK2512
R1 3m 1% WSK2512
Q1 SQM120P10_10M1L
Q2 SQM120P10_10M1L
R2 10k
D1 12V BZT52C12
Q3 MMBTA42 (300 V)
R3 1.2k
R4 1k
R5 100k
R6 499k
R7 499k
R8 33k 0.1%
C1 100n
U1 INA241A2xD
C2 100n
R9 1k
C3 100n
D2 ES1D (200 V 1 A)

BAT2

BAT2 channel: 1 mOhm Kelvin shunt, INA241A2 (20 V/V, REF=Vs/2 -> bidirectional, +-82 A FS), back-to-back P-FET switch. Gate driver sinks ~2 mA via MMBTA42. 12 V zener sets Vgs independent of bus voltage (105..205).

J2 BAT2+ (M4 ring) WSK2512
R10 3m 1% WSK2512
Q4 SQM120P10_10M1L
Q5 SQM120P10_10M1L
R11 10k
D3 12V BZT52C12
Q6 MMBTA42 (300 V)
R12 1.2k
R13 1k
R14 100k
R15 499k
R16 499k
R17 33k 0.1%
C4 100n
U2 INA241A2xD
C5 100n
R18 1k
C6 100n
D4 ES1D (200 V 1 A)

BAT3

BAT3 channel: 1 mOhm Kelvin shunt, INA241A2 (20 V/V, REF=Vs/2 -> bidirectional, +-82 A FS), back-to-back P-FET switch. Gate driver sinks ~2 mA via MMBTA42. 12 V zener sets Vgs independent of bus voltage (105..205).

J3 BAT3+ (M4 ring) WSK2512
R19 3m 1% WSK2512
Q7 SQM120P10_10M1L
Q8 SQM120P10_10M1L
R20 10k
D5 12V BZT52C12
Q9 MMBTA42 (300 V)
R21 1.2k
R22 1k
R23 100k
R24 499k
R25 499k
R26 33k 0.1%
C7 100n
U3 INA241A2xD
C8 100n
R27 1k
C9 100n
D6 ES1D (200 V 1 A)

BAT4

BAT4 channel: 1 mOhm Kelvin shunt, INA241A2 (20 V/V, REF=Vs/2 -> bidirectional, +-82 A FS), back-to-back P-FET switch. Gate driver sinks ~2 mA via MMBTA42. 12 V zener sets Vgs independent of bus voltage (105..205).

J4 BAT4+ (M4 ring) WSK2512
R28 3m 1% WSK2512
Q10 SQM120P10_10M1L
Q11 SQM120P10_10M1L
R29 10k
D7 12V BZT52C12
Q12 MMBTA42 (300 V)
R30 1.2k
R31 1k
R32 100k
R33 499k
R34 499k
R35 33k 0.1%
C10 100n
U4 INA241A2xD
C11 100n
R36 1k
C12 100n
D8 ES1D (200 V 1 A)

PV

PV channel: 1 mOhm Kelvin shunt, INA241A2 (20 V/V, REF=Vs/2 -> bidirectional, +-82 A FS), back-to-back P-FET switch. Gate driver sinks ~2 mA via MMBTA42. 12 V zener sets Vgs independent of bus voltage (105..205).

J5 PV+ (M4 ring) WSK2512
R37 1m 1% WSK2512
Q13 SQM120P10_10M1L
Q14 SQM120P10_10M1L
R38 10k
D9 12V BZT52C12
Q15 MMBTA42 (300 V)
R39 1.2k
R40 1k
R41 100k
R42 499k
R43 499k
R44 33k 0.1%
C13 100n
U5 INA241A2xD
C14 100n
R45 1k
C15 100n
D10 ES1D (200 V 1 A)
D11 SMCJ58A (TVS, reverse-PV clamp)

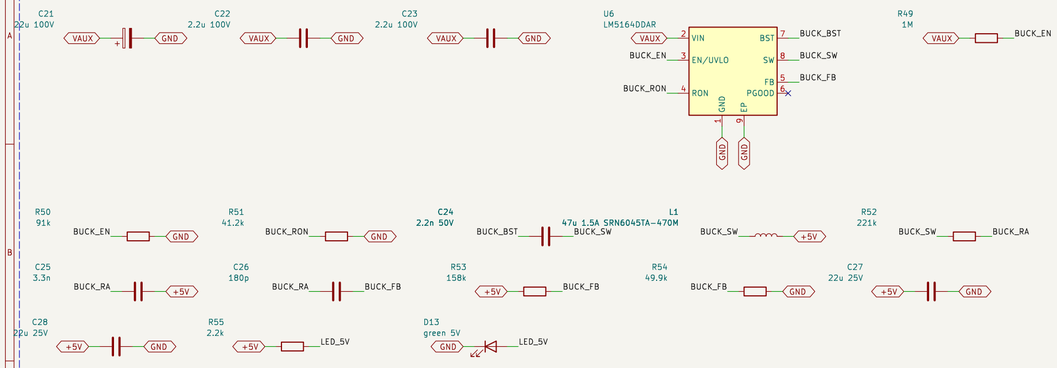
BUS

Common bus: lugs, bulk capacitance, TVS, bus voltage divider. GND lugs take pack negatives, PV- and load returns.

J6 BUS+ (M4 ring)
J7 BUS+ (M4 ring)
J8 GND (M4 ring)
J9 GND (M4 ring)
J10 GND (M4 ring)
C16 100u 100V
C17 100u 100V
C18 1u 100V
C19 1u 100V
C20 100n
D12 SMCJ58A (TVS)
R46 499k
R47 499k
R48 33k 0.1%

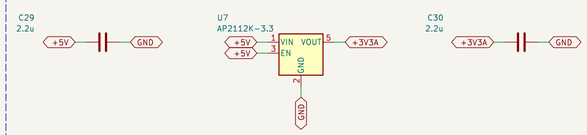
BUCK

LM5164 COT buck VAUX(18..100 V) -> 5 V 1 A, 300 kHz. RON=5*2500/300=41.2k. Type-3 ripple: RA*CA <= ton*(Vin-Vout)/20mV -> 221k/3.3n (19–21 mV @ 30–84 V). CB>=75us/(3*RFB1). UVLO on at 18 V.



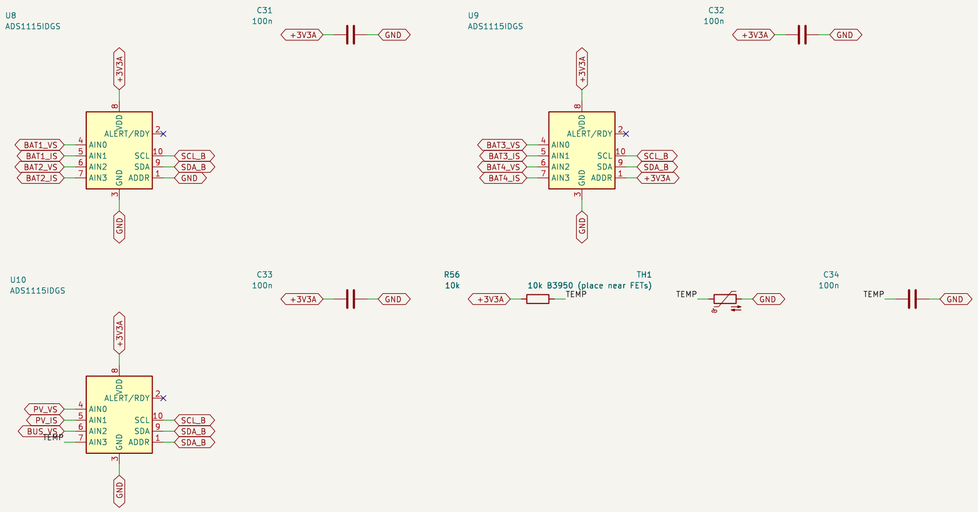
LDO

AP2112K-3.3: analog / I2C rail +3V3A.



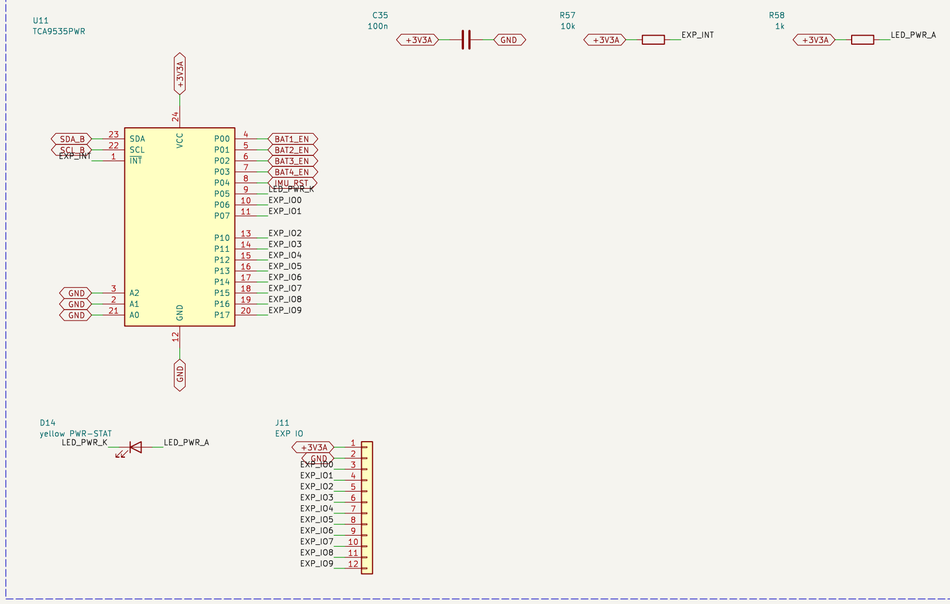
PADC

Power ADCs on I2C bus B (ADS1115 @3.3 V, PGA +-4.096 V). 0x48: BAT1/BAT2, 0x49: BAT3/BAT4, 0x4A: PV, BUS, board temp.



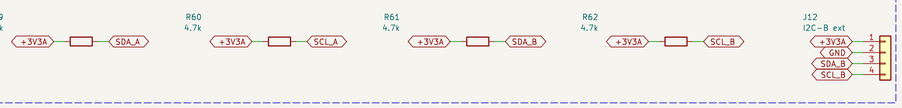
EXP

TCA9535 @0x20 (bus B). No internal pull-ups -> pack switches default OFF at power-up; state survives ESP32 reset/OTA.



I2C

I2C pull-ups (4.7k to +3V3A) and bus B expansion connector.



Sheet: supply_sense
Main controller — ESP32-S3
solbot
Sheet: /supply_sense/
File: supply_sense.kicad_sch

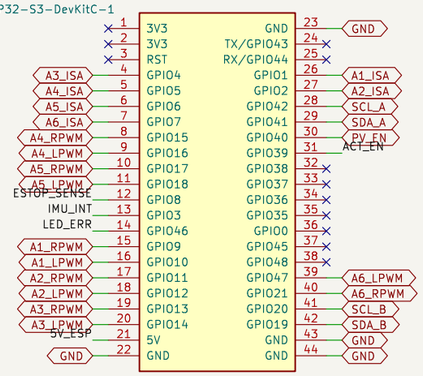
Title: Logic supply, power ADCs, IO expander

Size: A2	Date: 2026-10-15	**Rev: v4.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

ESP

ESP32-S3-DevKitC-1 (N8/N8R2) in 2x22 sockets. Program & log via the "UART" USB port only; GPIO19/20 (USB-OTG) are used as I2C bus B. GPIO35-37 left free (octal PSRAM on R8 variants), GPIO38/48 = on-board RGB LED.

U12
ESP32-S3-DevKitC-1



D15
SS34 (blocks USB back-feed)



C36
10u



R63
1k



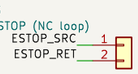
D16
red ERR



ESTOP

Hardware E-stop: ACT_EN (GPIO39) is looped through the NC E-stop contact; loop open or ESP not driving -> 74HCT244 outputs Hi-Z -> 10k pull-downs -> all IBT-2 PWM/EN low. Fit a wire bridge if no E-stop is used.

J13
E-STOP (NC loop)



R64
470



R65
1k



R66
100k



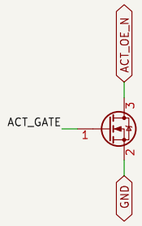
C37
100n



R67
10k



Q16
2N7002



R68
10k



IMU

BNO085 (Adafruit 4754 breakout) on cable. 5 V supply, I2C bus A @0x4A, INT -> GPIO3 (not a boot strap unless eFuse set), RST via TCA9535.

J14
IMU BNO085



R69
10k



C38
10u



MECH

Mounting holes M3.



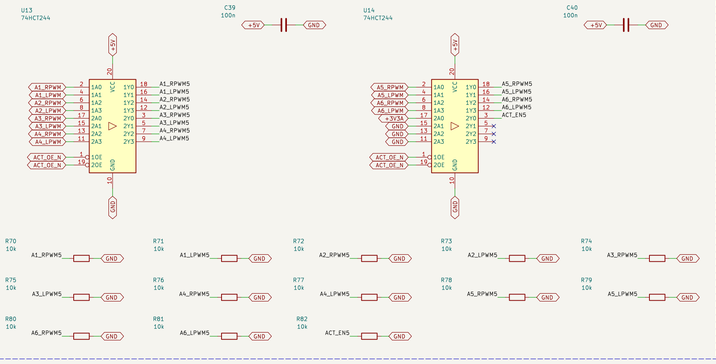
Sheet: mcu
Main controller — ESP32-S3
solbot
Sheet: /mcu/
File: mcu.kicad_sch

Title: ESP32-S3, IMU, E-stop

Size: A3	Date: 2026-10-15	**Rev: v4.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

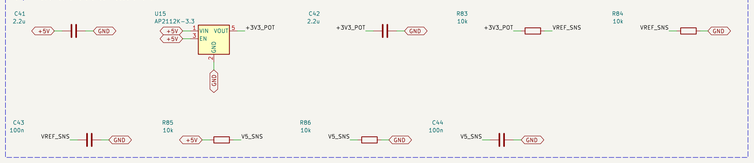
BUF

74HCT244 level shifters 3.3 V -> 5 V for IBT-2 inputs. /OE driven by E-stop gate. 10k pull-downs keep IBT-2 inputs low when Hi-Z.



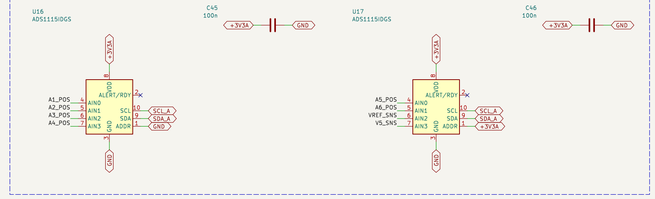
POSREF

Pot excitation rail (separate AP2112 so a shorted sensor cable cannot take down the ADC/I2C rail) + ratiometric reference measurement.



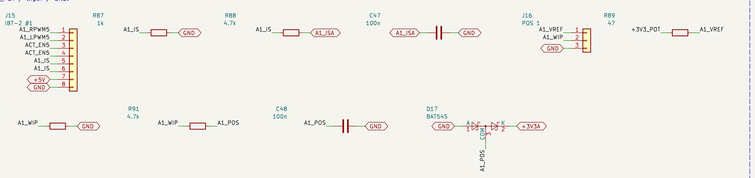
AADC

Actuator position ADCs on I2C bus A: 0x48 = A1…A4, 0x49 = A5, A6, VREF/2, 5V/2.



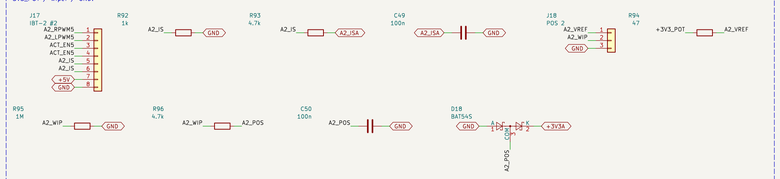
A1

Actuator 1 (steering). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



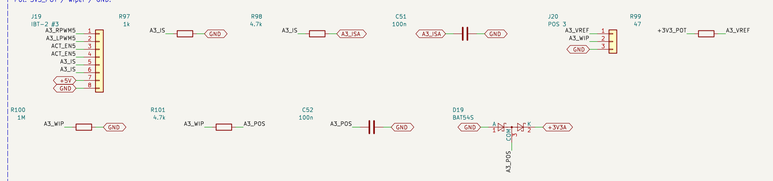
A2

Actuator 2 (tool lift). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



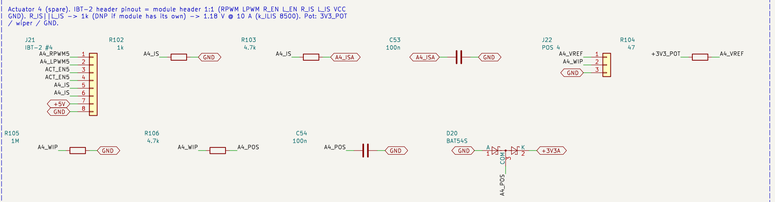
A3

Actuator 3 (tool slider). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



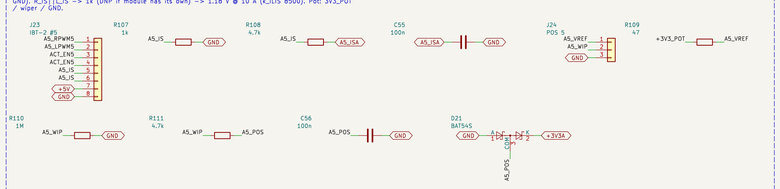
A4

Actuator 4 (spare). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



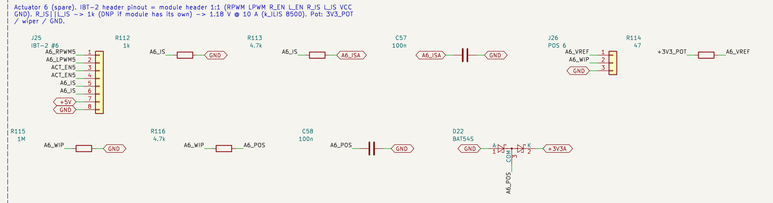
A5

Actuator 5 (spare). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



A6

Actuator 6 (spare). IBT-2 header pinout = module header 1:1 (RPWM LPWM R_EN L_EN R_IS L_IS VCC GND). R_IS||L_IS -> 1k (DNP if module has its own) -> 1.18 V @ 10 A (k_ILIS 8500). Pot: 3V3_POT / wiper / GND.



Sheet: actuators
Main controller — ESP32-S3
solbot

Sheet: /actuators/
File: actuators.kicad_sch

Title: Linear actuators: 6x IBT-2, position & current sense

Size: A1	Date: 2026-10-15	**Rev: v4.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/5